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Remagnetization and northward translation of Mesozoic red chert from Cedros Island and the San Benito Islands, Baja California, Mexico

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ABSTRACT

Samples for paleomagnetic analysis were collected from red radiolarian ribbon chert in a Franciscan-like subduction complex (Western Baja terrane) exposed along the western margin of the Baja California Peninsula. A 40-m-thick section of chert was sampled on Cedros Island, and a 30-m-thick and two smaller sections of chert were sampled on the San Benito Islands to the west. A site in pillow basalt underlying the chert also was sampled on Cedros Island. The sections of chert on Cedros Island and the San Benito Islands contain radiolaria ranging in age from Late Triassic to Early Cretaceous. All of these rocks have similar normal-polarity direction of magnetization, indicating that they were remagnetized long after deposition. Results from a chert site, including an intrastratal fold, also indicate that the magnetization is postdepositional. A significantly improved grouping (95% confidence level) of directions after correction for the structural attitudes of the Cedros and San Benito chert sections, however, implies that the remagnetization predates breaking up and incorporation of the chert into the subduction complex. Studies of modern subduction zones imply that the attitude of the ancient subducting slab, including the chert sections, was near horizontal (dip angles $<10°$) at the time of remagnetization. Thus, assuming paleohorizontality, a paleolatitude of $12° \pm 5°$ N can be estimated for the Western Baja terrane in mid-Cretaceous time. The paleomagnetic directions also indicate a subsequent poleward translation of $25° \pm 7°$ in latitude (2,800 km) and a clockwise rotation of $56° \pm 7°$ of the Western Baja terrane with respect to North America. The paleolatitude error limits at the 95% confidence level represent minimum values, because deviations of the bedding planes from horizontal during remagnetization would contribute to the error. A 10° NE dip of the subducting slab, however, would account for only 4° of the paleolatitude anomaly, and a $\simeq 60°$ NE dip would be needed to account for the entire anomaly. Modern analogs, in combination with plate-reconstruction models, imply that the northward translation of the Western Baja terrane along the ancient convergent margin of North America could have been accomplished by arc-parallel strike-slip faulting that was driven by oblique subduction.

INTRODUCTION

Subduction-complex rocks similar to those of the Franciscan Complex in California are exposed to the south in Mexico along the western coast of the Baja California Peninsula (Cohen and others, 1963; Jones and others, 1976; Kilmer, 1979). Taken together, these rocks record an extensive history of plate convergence along much of the western margin of ancestral North America during Late Jurassic through early Tertiary time, in part by showing varying degrees of blueschist-facies metamorphism (Evans and Brown, 1986). Incorporated within the subduction complex are blocks and slabs of pillow basalt and radiolarian chert, or basalt and pelagic limestone, that were accreted from one or more subducting oceanic plates (Blake, 1984). Paleomagnetic investigations of such fragments of oceanic crust hold the promise of providing important information on late Mesozoic to early Cenozoic plate motions that bear on the tectonic development of western North America (Atwater, 1970).

Because of the complicated metamorphic and tectonic history of Franciscan rocks, few studies have concluded that they retain their original remanent magnetizations. Such studies include that of Gromme and Gluskoter (1965) on Lower Cretaceous pillow lava and diabase in the Nicasio Reservoir terrane north of San Francisco (reanalyzed by Gromme, 1984), which concluded that these rocks were deposited at 28°N paleolatitude and were subsequently translated 20° in latitude northward and rotated 106° clockwise relative to cratonic North America. Alvarez and others (1980) determined that blocks of the mid-Cretaceous Laytonville Limestone in northern California originated at 17°S paleolatitude; using similar techniques, Courtillot and others (1985) determined that the approximately coeval Calera Limestone exposed to the south of San Francisco was initially deposited at 24°N paleolatitude. Tarduno and others (1985, 1986) independently confirmed the conclusions of this previous body of work on Franciscan limestones. These paleomagnetic data imply that the oceanic rocks of the Franciscan Complex originated far to the south of their present positions with respect to North America. This inference is consistent with plate-motion models that predict significant northward tangential components of Farallon–North America and especially Kula–North America relative motion during Cretaceous and early Tertiary time (Engebretson and others, 1985; Stock and Molnar, 1988). The uncertainty in the position of the Kula-Farallon ridge during this time, however, prevents precise determination of the history of tangential motion at a particular latitude along the western margin of North America (Stock and Molnar, 1988).

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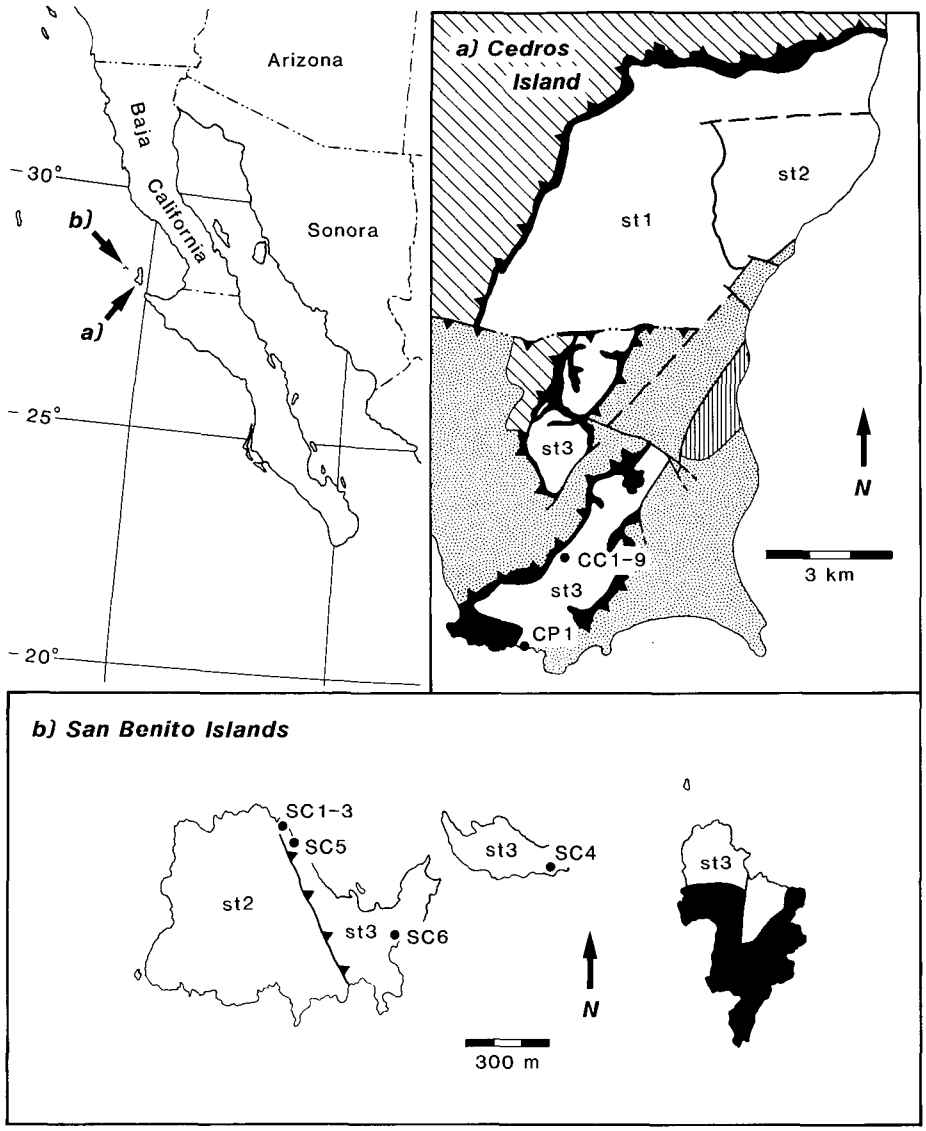


Figure 1. Location map of Cedros Island and the San Benito Islands off the western coast of Baja California, Mexico. Geologic maps (a) of southeastern Cedros Island and (b) of the San Benito Islands, showing paleomagnetic sampling sites; subterranes 1, 2, and 3 (st1, st2, st3; see text) of the Western Baja terrane (no pattern); upper-plate rocks of the Choyal terrane (diagonal lines); and overlying Upper Cretaceous Valle Formation (stippled pattern) and Miocene and Pliocene units (vertical lines). Fault-bounded serpentinite-matrix mélange is black; heavy lines indicate fault contacts (teeth on upper plate); lighter lines indicate depositional contacts (after Sedlock, 1988a, 1988b).

In order to determine the geographic origins of the protoliths of a Franciscan-like subduction complex in western Mexico, and perhaps contribute to the understanding of plate interactions along the western margin of North America, we present in this study paleomagnetic directions for samples of red radiolarian ribbon chert from Cedros Island and the San Benito Islands of Baja California (Fig. 1). Radiolarian chert is an essential constituent of layer 1 of the oceanic crust and is a common, although volumetrically minor, constituent of ancient subduction complexes. Aside from its clear association with oceanic plates, red chert with a stable remanent magnetization brings several potential advantages to a paleomagnetic study. First, chert commonly contains radiolarian biota which give direct estimates of the time of deposition for the units sampled. Acquisition of the characteristic magnetization, however, may or may not correspond with the time of deposition (Shibuya and Sasajima, 1986). Second, bedding planes in chert units are usually clearly defined by thin shale partings so that accurate tilt corrections can be made. Third, coherent sequences of chert usually represent enough time to average secular variation and to record reversals of the ancient geomagnetic field. The main disadvantage, however, is that metamorphism of the subduction complex may have remagnetized the rocks contained within. Although the cherts of Cedros Island and the San Benito Islands are shown in this paper to have been remagnetized, probably in Early Cretaceous time, we infer that they were remagnetized while at shallow dip angles, and suggest that their remanent magnetizations can be used to document subsequent tectonic displacements.

REGIONAL SETTING

Mesozoic rocks in the west-central region of the Baja California Peninsula are divided into three shallowly dipping, fault-bounded units: an upper plate of Triassic to Cretaceous, arc-related rocks; a lower plate of blueschist-facies, subduction-complex rocks that is strikingly similar to parts of the Franciscan Complex of California; and an intervening serpentinite-matrix mélange (Sedlock, 1988a, 1988b). This organizational scheme, as well as the identification of tectonostratigraphic terranes in the area by Sedlock (1988b), supersedes that of Kimbrough (1985).

The upper plate is a composite terrane consisting of (1) the Choyal terrane on Cedros Island, which is a Middle Jurassic arc-ophiolite complex overlain by Upper Jurassic volcanic and volcaniclastic rocks (Kilmer, 1979; Kimbrough, 1985; Baldwin and others, 1987); (2) the Vizcaino Norte and Vizcaino Sur terranes on the Vizcaino Peninsula, which contain Upper Triassic ophiolite overlain by different sequences of possibly related Jurassic and Cretaceous magmatic and sedimentary rocks (Moore, 1985); and (3) mid-Cretaceous flysch that overlaps the older rocks (Barnes, 1984). The upper-plate rocks are separated from the Cretaceous volcanic and plutonic core of the Baja California Peninsula by the Vizcaino Basin, which contains Upper Cretaceous to Quaternary sediments (Kimbrough, 1985) that obscure the probable fault-bounded eastern limit of Mesozoic oceanic basement rocks (Coney and others, 1980).

The upper plate is separated from blueschists of the lower plate by a serpentinite-matrix mélange which consists of blocks of eclogite,

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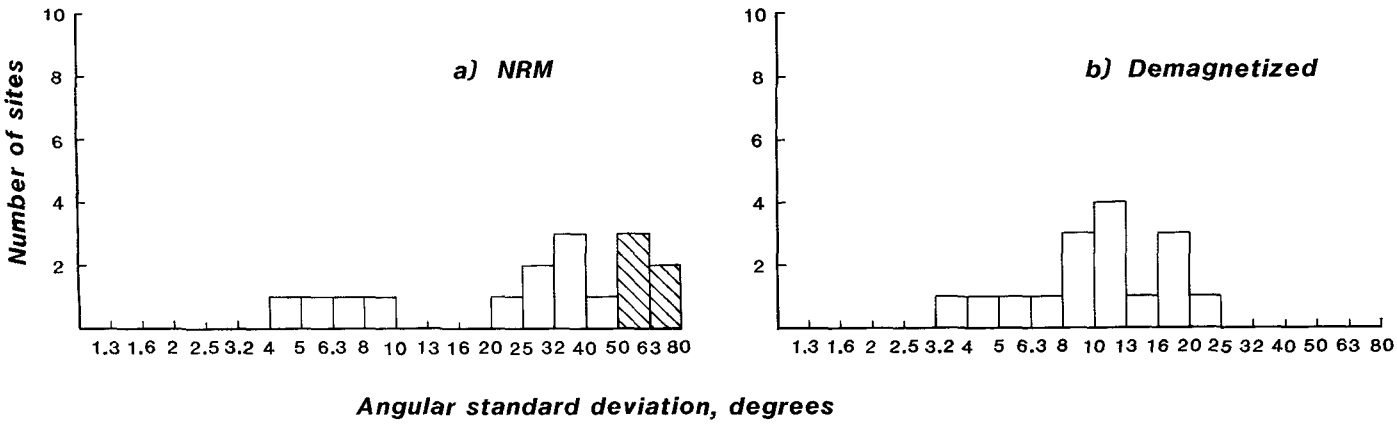


Figure 2. Histograms showing reduction in angular standard deviation of magnetization directions for all sites as a result of AF and thermal demagnetization. NRM is natural remanent magnetization as was first measured, and shaded columns indicate randomly distributed directions (Watson, 1956).

coarse-grained blueschist, amphibolite, epidote-amphibolite, and variably serpentinized mafic and ultramafic rocks enclosed in a strongly foliated lizardite-chrysotile matrix. This mélange is exposed on San Benito East Island (Cohen and others, 1963), in the southern half of Cedros Island (Kilmer, 1979; Sedlock, 1988b), and in a small outlier on the Vizcaino Peninsula near Puerto Nuevo (Moore, 1986). All blueschists of the lower plate are assigned to the Western Baja terrane because of their gross lithologic, structural, and metamorphic similarities (Sedlock, 1988b). The Western Baja terrane is subdivided into three fault-bounded subterranes on the basis of variations in the intensity of fabric development, intensity of recrystallization, mineral assemblages and compositions, and inferred P-T history (Fig. 1; Sedlock, 1988a, 1988b). Peak metamorphic conditions for the subterranes are >8 kbar, 225–235 °C for subterrane 1; 7–8 kbar, 170 °C for subterrane 2; and 5–6 kbar, 175–200 °C for subterrane 3 (Sedlock, 1988b). Within each subterrane, slip on a complex network of faults has disrupted an originally coherent stratigraphic sequence of interbedded oceanic rocks which includes mafic volcanic rocks, siliciclastic turbidite, chert, and limestone. Thus, the subterranes presently consist of complexly juxtaposed fault-bounded packets which contain remnants of the original stratigraphic sequence, tens to hundreds of meters thick, that are laterally traceable for as much as 3 km.

This study concerns rocks of subterrane 3, the least deformed and recrystallized of the subterranes. The thickness of the original stratigraphic sequence within subterrane 3 cannot be precisely determined because of difficulties in correlating rocks between different fault-bounded packets. Stratigraphically lowest in subterrane 3, there is a sequence of ocean-floor basalts which is at least several hundred meters thick. This sequence includes pillow basalt, massive flows, volcaniclastic breccia, and thin (<2-m) interbeds of limestone and chert. These volcanic rocks are depositionally overlain by a thick sequence of red radiolarian ribbon chert. On southern Cedros Island, this sequence is 40 m thick and uncontaminated by clastic debris. On the San Benito Islands, the sequence is about 30 m thick and is interbedded with coarse-grained mafic volcaniclastic rocks (Sedlock and Isozaki, 1990). Radiolarian biostratigraphy indicates that the chert ranges in age from Late Triassic to late Early Cretaceous (Sedlock and Isozaki, 1990). The chert is depositionally overlain by a sequence of thin-bedded siliciclastic turbidite that is at least 250 m thick. The protoliths of this subduction complex are interpreted as Late Triassic oceanic crust that was covered by siliciclastic turbidite as the oceanic plate neared a continental margin in late Early Cretaceous time (Sedlock, 1988b). $^{40}Ar/^{39}Ar$ plateau ages from lower-plate rocks indicate that the peak of blueschist-facies metamorphism occurred at about 110 Ma (Baldwin and Harrison, 1989; S. L. Baldwin, unpub. data) and uplift ages of subterrane 3 rocks are as old as 75 Ma (Baldwin and Harrison, 1989). These radiometric ages are similar to Early Cretaceous ages reported from the coherent schists of the Eastern Belt of the Franciscan Complex of California (for summary, see Cloos, 1986).

Paleomagnetic data for Cretaceous igneous and sedimentary rocks from the Baja California Peninsula (Teissere and Beck, 1973; Hagstrum and others, 1985; Filmer and Kirschvink, 1989) indicate that the Baja California Peninsula of Mexico and southern California has moved northward ~9° in latitude and rotated ~24° clockwise with respect to cratonic North America between Late Cretaceous time and the Neogene opening of the Gulf of California. These data include results for samples of the Valle Formation (Patterson, 1984; Hagstrum and others, 1985), a thick (10-km) sequence of mid-Cretaceous turbidite, which unconformably overlies the upper plate of the Mesozoic basement rocks on Cedros Island and the Vizcaino Peninsula. Because the lower-plate subduction-complex rocks are in fault contact with the upper-plate rocks, no constraint can be placed on the time at which the subduction-complex rocks were attached to the Cretaceous rocks of the Baja Peninsula. Northward movement of the Baja California Peninsula between Late Cretaceous and Miocene time (Hagstrum and others, 1987) is inferred to have occurred along transcurrent faults and to have been driven by oblique subduction (see also Beck, 1983, 1986).

FIELD AND LABORATORY PROCEDURES

Samples were initially collected from a variety of units in the subduction-related basement rocks exposed on Cedros Island to determine which were most suitable for paleomagnetic investigation. These rocks were expected to retain primary magnetizations because the peak temperatures and duration of blueschist metamorphism (170–200 °C, ≈35 Ma) are well below the time-temperature maxima for thermal remagnetization of the constituent magnetic minerals, that is, magnetite and hematite (Pullaiah and others, 1975). Ubiquitous quartz ± carbonate veins in the subduction-complex rocks, however, are a clear indication of prevalent

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chemical alteration associated with the high-pressure, low-temperature metamorphism. Preliminary results from fine-grained sandstones, pillow basalts, and red radiolarian chert showed that only the basalt and chert contained a stable remanent magnetization. Because paleohorizontal was difficult to determine in the pillow basalt and because of the aforementioned potential advantages of red radiolarian chert, subsequent sampling was limited to coherent sections of chert on Cedros Island and the San Benito Islands to the west.

A 40-m-thick section of chert on Cedros Island (Fig. 1) was sampled at nine sites (CC1–9). Samples were collected at 0.5- to 2-m stratigraphic intervals in an attempt to determine an Early Jurassic to Early Cretaceous polarity stratigraphy and to check for any progressive change in inclination that might indicate tectonic transport. In addition, samples were collected from around an intrastratal fold at site CC1. Pillow basalt underlying the chert was sampled at site CP1. On the San Benito West Island, four sites (SC1–4) were collected from a 30-m-thick section of Late Jurassic age in order to obtain both normally and reversely magnetized samples. Partial chert sequences in smaller fault-bounded packets that had different attitudes also were sampled in order to determine the timing of magnetization with respect to breakup of the original stratigraphic sequence of subterrane 3. Site SC5 was collected from Early Jurassic chert on San Benito Central Island, and SC6 was collected from chert of unknown but presumed Jurassic age on San Benito West Island.

Oriented samples were collected at sites CC1, CC2, and CP1 as 2.5-cm drill cores using a portable gasoline-powered drill. The remaining chert samples were collected as hand samples due to the extreme durability of the rock. Solar and magnetic compasses and a clinometer were used to measure the orientations of the drill cores, and a magnetic compass and clinometer were used to orient the hand samples. Accuracy of orientation for the samples is estimated to be $\simeq 2°$ for field-drilled cores and $\simeq 5°$ for hand samples. In the laboratory, 2.5-cm-diameter cores were drilled from the hand samples, and specimens ~2.2 cm in length were cut from the cores. Specimens were subjected to either alternating-field (AF) demagnetization in peak fields up to 100 mT or thermal demagnetization to temperatures exceeding 675 °C. In addition, progressive acquisition of isothermal remanent magnetization (IRM) in fields up to 0.7 T was measured on selected small cores to aid in the identification of the magnetic mineralogy. The remanent magnetization of individual specimens was measured with a cryogenic magnetometer. The direction of magnetization for each sample was determined from the demagnetization data using a least-squares method based on principal-component analysis (Kirschvink, 1980), and the statistical methods of Fisher (1953) were used to analyze the within-site line directions and overall averages. Structural corrections were made by a simple rotation of the bedding plane to horizontal about the strike axis.

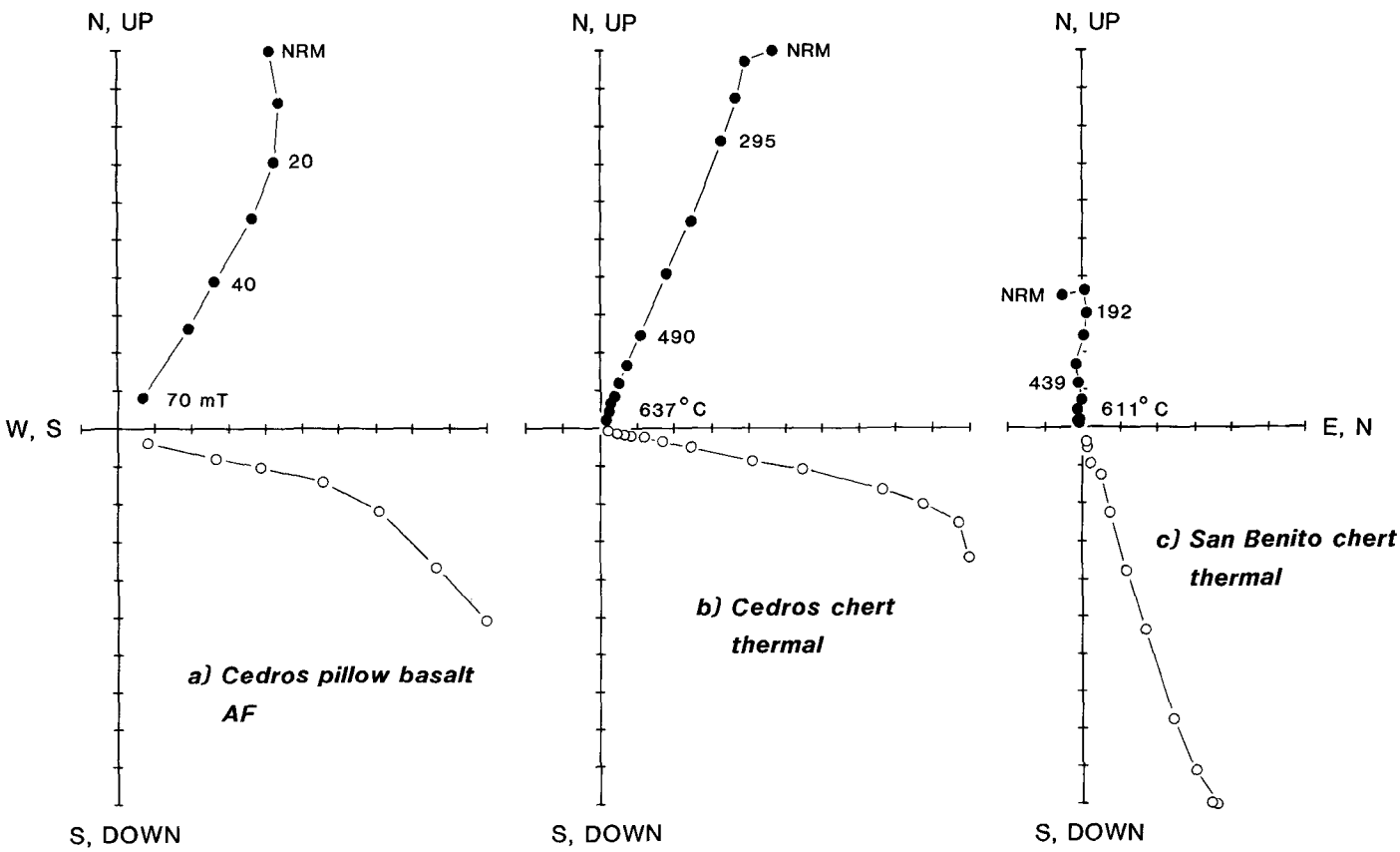


Figure 3. Orthogonal projections of uncorrected demagnetization vector endpoints for samples of (a) Cedros pillow basalt, (b) Cedros chert, and (c) San Benito chert. AF denotes alternating field demagnetizations, and thermal denotes thermal demagnetizations. Solid symbols indicate projections of vector endpoints onto the horizontal plane; open symbols, onto the vertical plane.

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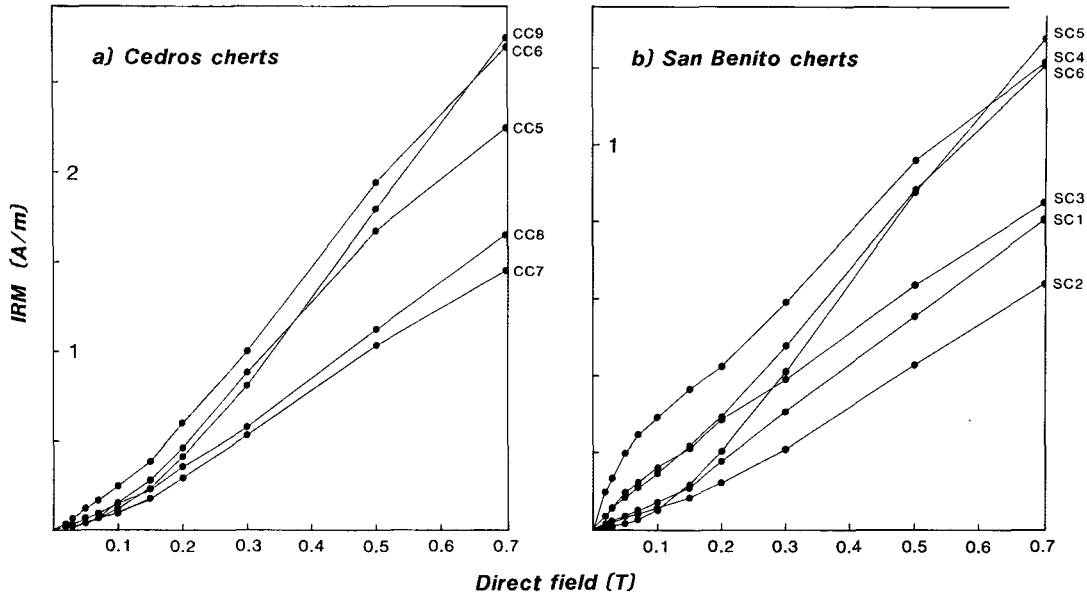


Figure 4. Isothermal remanent magnetization acquisition curves for samples of red chert from (a) Cedros Island and (b) the San Benito Islands.

RESULTS

Measurements of intensities and directions of natural remanent magnetization (NRM) were made on all samples from Cedros Island and the San Benito Islands prior to demagnetization treatments. NRM intensities for the chert samples are relatively uniform, and site averages range from 5.09×10^{-3} A m^{-1} to 2.44×10^{-2} A m^{-1}. The average NRM intensity for the site in pillow basalt is 2.42×10^{-3} A m^{-1}. The dispersion of NRM directions for each site, however, ranges from being tightly grouped to being highly scattered (Fig. 2). NRM directions for sites CC1, CC2, CC3, and CP1 have angular standard deviations of less than 10°; sites CC8, CC9, SC1, SC3, and SC6 have randomly distributed directions (Watson, 1956); the remaining sites have angular standard deviations between 10° and 45°. The scattering of the NRM directions is most likely caused by isothermal remanent magnetizations (IRM) related to direct or nearby lightning strikes at the sampling localities (Graham, 1961; Cox, 1961).

Both AF and thermal treatments were successful in demagnetizing the characteristic component of magnetization for samples of pillow basalt collected beneath the chert section on Cedros Island. AF demagnetization removed what appeared to be a recent viscous overprint from the basalt samples, isolating the characteristic component between 30 and 100 mT (Fig. 3a). The same component was isolated by thermal demagnetization between 200 and 500 °C. Growth of a new magnetic phase was observed in the basalt samples after they were heated above 500 °C. AF demagnetization to 100 mT generally reduced the chert samples' NRM intensities to about one-half of their initial values. A component of magnetization with inconsistent directions, probably related to lightning strikes, was commonly removed by 20 to 50 mT, and either a stable endpoint was reached or a single component of magnetization was removed above 50 mT. Thermal demagnetization was successful in completely demagnetizing the chert samples and recovering a consistent characteristic component of magnetization (Figs. 3b, 3c). Randomly oriented secondary components, where present, were usually removed by heating to 500 °C, and the characteristic component was most often removed by 620 to 650 °C.

IRM acquisition experiments were performed on representative small specimens from most of the chert sites. The IRM curves for specimens from Cedros Island (Fig. 4a) indicate that hematite is the dominant remanence carrier and

TABLE 1. SITE-MEAN DIRECTIONS AND POLES

Site	Geographic		Str/Dp	Stratigraphic		N	R	k	α_{95}	Pole	
	I	D		I	D					Lat.	Long.
Cedros chert and pillow basalt											
CC1	9.3	28.3	140/20	27.7	25.6	6	5.9885	436	3.2	62.8	358.7
CC2	7.9	0.8	140/20	20.5	356.8	6	5.9377	80	7.5	72.3	75.3
CC3	20.8	37.7	140/20	40.2	34.8	6	5.9760	208	4.7	58.2	336.1
CC4	13.5	29.4	140/20	32.0	26.2	6	5.8405	31	12.2	63.7	353.1
CC5	9.6	30.0	140/20	28.2	27.5	8	7.8672	53	7.7	61.4	356.2
CC6	15.8	22.8	140/20	33.3	18.3	8	7.6924	23	11.9	70.5	1.3
CC7	11.7	31.0	140/20	30.5	28.3	7	6.5796	14	16.5	61.4	352.8
CC8	12.9	10.9	140/20	28.0	5.9	5	4.9643	112	7.3	75.7	41.0
CC9	11.8	20.4	140/20	28.9	16.4	4	3.9680	94	9.5	70.2	11.1
CP1	11.6	31.2	140/20	30.4	28.5	8	7.9748	278	3.3	61.3	352.7
San Benito chert											
SC1	44.9	12.6	1/73	4.0	46.9	7	6.9080	65	7.5	38.1	356.3
SC2	50.8	2.4	347/74	3.0	39.4	5	4.9447	72	9.1	43.8	2.8
SC3	53.7	353.8	4/79	14.9	56.9	7	6.8843	52	8.5	32.6	344.1
SC4	54.1	356.9	356/70	15.6	48.6	7	6.7224	22	13.3	40.0	348.7
SC5	-5.9	53.9	157/36	29.1	52.0	7	6.8508	40	9.6	40.5	337.8
SC6	70.2	325.1	332/99	-6.2	42.2	5	4.8178	22	16.7	38.7	5.1
Averages											
	25.2	20.9				16	14.3163	9	13.1	65.4	7.8
				23.4	31.4	16	15.0989	17	9.3	57.1	356.4

Note: I and D, inclination and declination of paleomagnetic direction, in degrees, before (Geographic) and after (Stratigraphic) correction for structural tilt; Str/Dp, strike and dip of rock unit in degrees; N, number of samples analyzed; R, vector sum of N unit vectors; k, precision parameter (Fisher, 1953) for directions (K for poles); α_{95}, radius of 95% confidence limit for directions (A_{95} for poles), in degrees; Lat. and Long., latitude and east longitude of virtual geomagnetic pole, in degrees.

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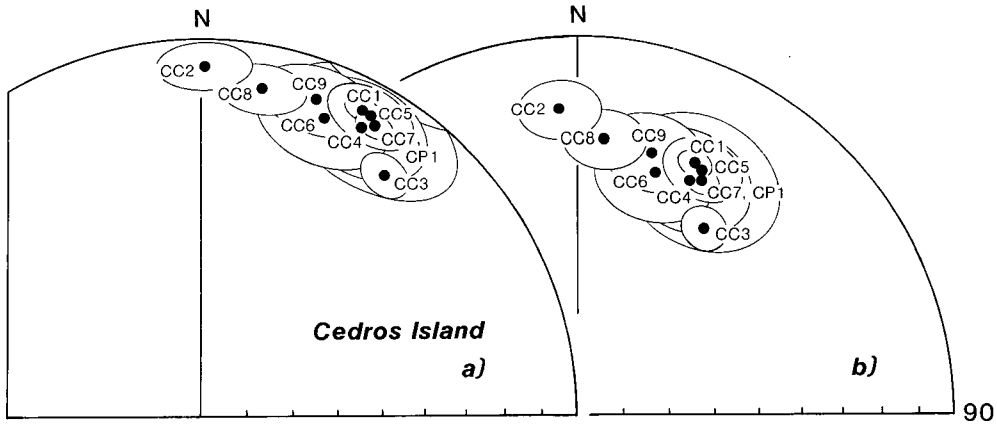


Figure 5. Mean directions of sites for Cedros Island (with 95% confidence limits) uncorrected (a) and corrected (b) for structural tilt (Table 1). Diagrams are equal-area plots where solid symbols indicate projection from the lower hemisphere.

that little if any magnetite is present in these chert samples. The more rapid increase in magnetization below 0.1 T for site CC6 implies that this specimen contains a small quantity of magnetite. IRM results for specimens from the San Benito Islands (Fig. 4b) are similar, but the slight convex-upward shape of the curves below 0.1 T for samples from sites SC3, SC4, and SC6 probably indicates the presence of small amounts of magnetite in these samples as well. These data in conjunction with the demagnetization data suggest that the characteristic component of magnetization in the cherts is carried by hematite with maximum blocking temperatures between 620 and 650 °C. The small amount of magnetite in the samples having lower intrinsic coercivity is more readily activated by lightning strikes and probably carries the random overprint components most easily removed with AF demagnetization.

TABLE 2. CRETACEOUS PALEOMAGNETIC POLES AND DISPLACEMENT PARAMETERS FOR THE CEDROS AND SAN BENITO CHERT AND BAJA PENINSULA RELATIVE TO NORTH AMERICA

Lat.	Long.	N	R	K	A_{95}	d	r	Paleolat.	Ref.
North America									
67.3	186.3	5	4.9921	505	3.4			36.7 ± 3.4	1
Baja Peninsula									
83.3	344.4	8	7.9734	263	3.4	11.0 ± 3.8	32.0 ± 4.5	26.7 ± 2.7	2
Cedros and San Benito chert									
57.1	356.4	16	15.2890	21	8.2	25.1 ± 6.9	55.7 ± 7.4	12.2 ± 5.3	3

Note: abbreviations as in Table 1. d, r, poleward displacement and clockwise rotation of site (28.1°N, 244.8°E) with respect to cratonic North America, in degrees, with 95% confidence limits after Beck (1980), as modified by Demarest (1983); Paleolat., paleolatitude with 95% confidence limits, in degrees, calculated for site; Ref., references: 1, Irving and Irving (1982, Table I, entries 39, 42–45); 2, Hagstrum and others (1985); 3, this study.

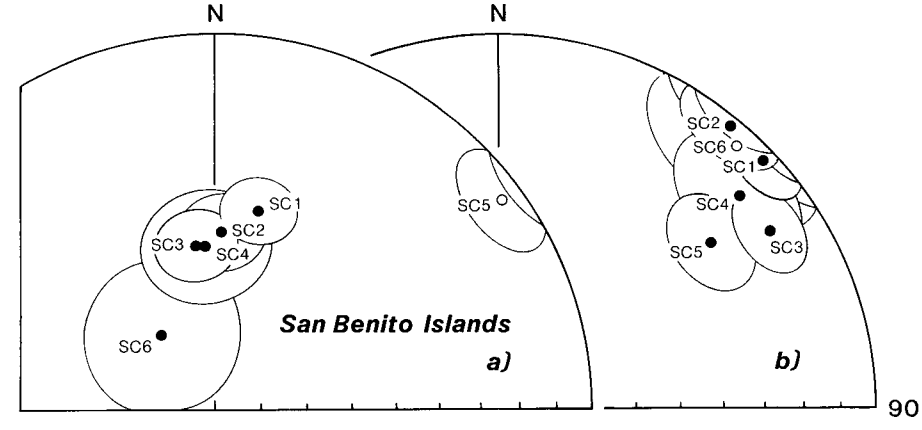


Figure 6. Mean directions of sites for the San Benito Islands (with 95% confidence limits) uncorrected (a) and corrected (b) for structural tilt (Table 1). Diagrams are portions of equal-area plots where solid symbols indicate projection from the lower hemisphere; open symbols, from the upper hemisphere.

Site-mean directions for the sections on Cedros Island and the San Benito Islands (Table 1) are shown both before and after application of structural corrections in Figures 5 and 6. Although the geomagnetic field was of mixed polarity during the Early Jurassic to Early Cretaceous deposition of the chert (Harland and others, 1982), all samples in this study are of normal polarity. This fact and the within-section similarity of directions indicate that the magnetization was acquired after deposition of the chert during an interval of normal polarity. The direction in the underlying pillow basalt is identical to that in the chert; evidently the basalts also were remagnetized at the same time. Dispersion of the characteristic directions for samples taken from around an intrastratal fold at site CC1 increases significantly (95% confidence level; Watson and Irving, 1957) after unfolding ($k_1 = 436$; $k_2 = 93$), showing that the remanent magnetization was most likely acquired after deposition and intrastratal deformation of the chert. For this reason, a single structural correction indicating the over-all orientation of the section was used for samples from Cedros Island. The dispersion of all sites on Cedros Island and the San Benito Islands, however, is significantly improved (95% confidence level; Watson and Irving, 1957) after structural correction ($k_1 = 9$, $k_2 = 17$), implying that the remagnetization predates the deformation represented by the between-site structural corrections.

REMAGNETIZATION OF SUBDUCTION COMPLEX CHERT

Red radiolarian chert of Mesozoic age has been collected for paleomagnetic study from subduction complexes exposed around the Pacific margin in Indonesia (Haile, 1978), Japan (Hattori, 1982; Shibuya and Sasajima, 1986), and western North America (Curry and others,

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1984; Hagstrum and others, 1985; this study). These cherts have been subjected to varying degrees of deformation and metamorphism associated with accretion to their respective land masses, and where sufficient data are available (Curry and others, 1984; Shibuya and Sasajima, 1986; this study), the cherts appear to have been partially or completely remagnetized.

Curry and others (1984) sampled a coherent section of pillow basalt, radiolarian chert, and quartzo-feldspathic sandstones and siltstones ranging in age from Early Jurassic to Late Cretaceous in the Marin Headlnds terrane just north of San Francisco, California. All samples of basalt, chert, and siltstone yield the same magnetization polarity and direction, indicating that the entire section was remagnetized during an interval of normal polarity, possibly the Cretaceous Normal Polarity Superchron (see Curry and others, 1984, p. 93). Because the strata are nearly homoclinal, fold tests for the basalt and chert are inconclusive; the siltstones pass a fold test at the 99% confidence level. IRM experiments show the dominant magnetic mineral to be hematite in the red chert samples and to be magnetite in the basalt and sandstone samples (Curry and others, 1984).

Shibuya and Sasajima (1986) collected samples of Middle Triassic to Lower Jurassic red chert and Middle Jurassic red siliceous mudstone from the limbs of a large-scale syncline in the Mino terrane of central Japan. They observed two characteristic components of magnetization in the chert samples using thermal demagnetization treatments, one between 380 and 550 °C and another between 550 and 670 °C. The higher blocking-temperature component was found only in samples from the southern limb. This component of magnetization passed an intrastratal fold test, conformed to stratigraphically related polarity intervals, and also passed a large-scale tectonic fold test. The higher blocking-temperature component was therefore inferred to have been acquired by the chert during deposition (Shibuya and Sasajima, 1986). The lower-blocking temperature component was the dominant component found in all samples. This component of magnetization failed the intraformational fold test and was only of normal polarity, but it passed the large-scale tectonic fold test. The lower blocking-temperature component was therefore considered by Shibuya and Sasajima (1986) to be a chemical remanent magnetization (CRM) acquired between deposition and tectonic folding of the chert, probably during accretion of the chert body. IRM experiments and blocking-temperature spectrum analysis showed the presence of both hematite and magnetite in the samples. Shibuya and Sasajima (1986) suggested that the lower blocking-temperature component was associated with magnetite and that the higher blocking-temperature component was associated with hematite.

Similar results to those of Curry and others (1984) and Shibuya and Sasajima (1986) were obtained in this study from chert within the subduction complex of the Western Baja terrane. Again, a single normal-polarity component of magnetization was found in a collection of samples that were initially deposited in both normal- and reversed-polarity geomagnetic fields. This magnetization is also shown to have been acquired after intrastratal folding of the chert. The apparent remagnetization of these rocks therefore must have occurred after deposition of the youngest chert (Early Cretaceous on Cedros Island, latest Early Cretaceous on San Benito West Island) but prior to disruption and incorporation of the original stratigraphic sequence into the subduction complex. These constraints, combined with the general correlation of remagnetization with reset K-Ar systems due to reheating or recrystallization, and with the radiometric ages discussed above, suggest that subterrane 3 rocks of Cedros Island and the San Benito Islands were remagnetized during subduction-related metamorphism at about 110 Ma (Baldwin and Harrison, 1989). A normal-polarity magnetization would be expected at this time because it is within the Cretaceous Normal Polarity Superchron. The remagnetization is most likely CRM and not a thermoremanent magnetization (TRM), because the peak metamorphic temperatures (<200 °C) are well below the blocking temperatures of the rock samples' magnetic minerals. Although Shibuya and Sasajima (1986) concluded that the remagnetization of the Mino terrane chert in Japan was associated with low-temperature growth of magnetite, our data imply that the characteristic magnetization for the chert of this study is carried by hematite and that magnetite is only a minor phase. It appears, however, that an equivalent CRM is carried by magnetite in pillow

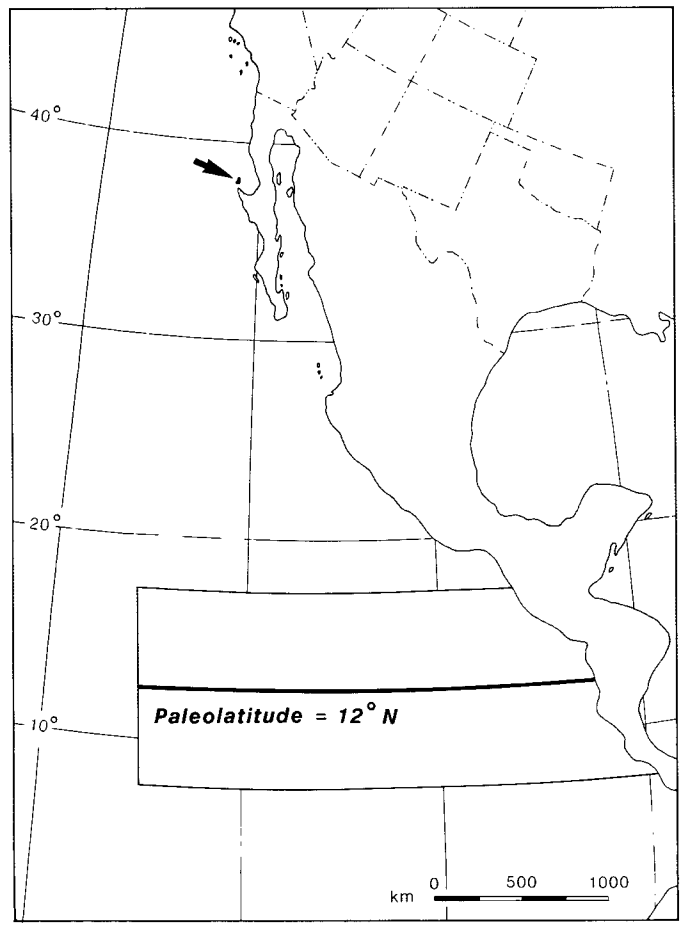


Figure 7. Equal-area map of southwestern North America for Cretaceous time, showing estimated paleolatitude, with 95% confidence limits (minimum value, see text), at which the chert of Cedros Island and the San Benito Islands (arrow) is inferred to have been subducted and remagnetized.

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TABLE 3. RECONSTRUCTION PARAMETERS FOR THE PACIFIC, FARALLON, AND KULA PLATES RELATIVE TO NORTH AMERICA AT 26°N, 248°E

Age (Ma)		Azimuth (degree)	Obliquity (degree)	Speed (km/m.y.)	Percentage of total convergence given to strike-slip displacement	Tangential displacement (km)
from	to					
0	5(P)	310	100	64	98	315
5	9(P)	297	113	61	92	224
9	17(P)	296	114	60	91	440
17	28(P)	298	112	56	93	572
28	37	52	23	86	15	116
37	43	56	19	128	15	115
43	48	62	13	163	15	122
48	56	56	19	160	15	192
56	61	60	15	140	15	105
56	61(K)	44	31	85	50	213
61	66	58	17	132	15	99
61	66(K)	33	42	106	50	265
66	74	64	14	112	15	134
66	74(K)	22	53	111	50	444
74	85	25	50	88	15	145
74	85(K)	348	87	122	50	671
85	100	38	37	99	15	223
100	110	62	13	61	15	92
					Total Pacific:	1,551
					Total Farallon:	1,343
					Total Farallon + Kula:	2,453

Note: all data are for Farallon plate except where (K) indicates data are for Kula plate and (P) for Pacific plate. Azimuth and speed for time intervals are from Engebretson and others (1985), Table 5, Baja California site (26°N, 248°E). Obliquity values are based on a 345°-trending North American plate margin in Late Cretaceous and Paleogene time and a 320°-trending margin in Neogene time (see text). Percentage of total convergence partitioned into strike-slip displacement for Farallon and Kula plates estimated from Jarrard (1986a, Table 5) after A. S. Jayko and M. C. Blake, Jr. (unpub. data) (see text).

basalt and clastic sediments under and above the red chert in the Marin Headlands terrane (Curry and others, 1984), as well as in the pillow basalt beneath the red chert in the Western Baja terrane.

TECTONIC IMPLICATIONS

Conclusions concerning the subsequent tectonic history of structurally deformed and remagnetized rocks are difficult to draw because attitudes of the rock units are generally unknown at the time of remagnetization. The positive "tilt" test in this study (95% confidence level), however, indicates that the characteristic magnetizations of the Cedros and San Benito chert sections predate most, if not all, of the tectonic deformation within the subduction complex. This result implies that these rocks were approximately coplanar and probably part of the subducting slab during remagnetization, prior to their incorporation within the overlying accretionary prism. The remagnetization of the chert sequences most likely accompanied late Early Cretaceous blueschist-facies metamorphism at the base of the accretionary prism. In many modern subduction zones, oceanic lithosphere is underthrust to depths corresponding to blueschist-facies conditions (15–30 km) along faults that dip less than 10° (Westbrook and others, 1982; White and Loudon, 1983; Platt and others, 1985; von Huene and others, 1987; Plafker, 1987). Where available, seismic reflection records suggest that sedimentary rocks atop subducted oceanic crust are underthrust tens of kilometers below the hanging wall without disruption (Westbrook, 1982; von Huene and others, 1987). We therefore infer that the rocks of subterrane 3 were metamorphosed and remagnetized while at shallow dip angles ($<10°$) within the subduction zone prior to underplating, disruption of the original stratigraphic sequence, and uplift of the subduction complex (<75 Ma).

Assuming that the bedding planes of the chert sections were horizontal at the time of remagnetization, a paleolatitude of $12° \pm 5°$ N can be estimated for the chert in mid-Cretaceous time (Fig. 7). This position ought to correspond to the approximate latitude at which the chert was subducted. The present position of the chert and its paleomagnetic directions, assuming paleohorizontality, also indicate a subsequent $25° \pm 7°$ northward displacement in latitude ($2{,}800 \pm 800$ km) and a clockwise rotation of $56° \pm 7°$ of these rocks relative to North America since mid-Cretaceous time (Table 2). The error limits at the 95% confidence level are *minimum* estimates, because any significant deviation of the bedding planes from horizontal during remagnetization would contribute to the error. A 10° northeastward dip of the subducting slab (see below) would account for only 4° of the paleolatitude anomaly, however, and an $\simeq 60°$ northeastward dip would be needed to account for the entire anomaly. The magnitude of the apparent northward translation and clockwise rotation of the Cedros and San Benito chert is similar to, but greater than, that indicated by paleomagnetic data for the Peninsular Ranges batholith and Upper Cretaceous strata of the Baja Peninsula (Hagstrum and others, 1985; Table 2). Modern analogs (Fitch, 1972; Jarrard, 1986a, 1986b) imply that the northward displacements of the Western Baja terrane and the Peninsular Ranges terrane along the ancient convergent margin of western North America could have been accomplished by oblique subduction and concurrent arc-parallel, strike-slip faulting (Beck, 1983, 1986).

Reconstructions of Farallon and Kula plate motions with respect to North America (Engebretson and others, 1985; Stock and Molnar, 1988) have shown that they were obliquely subducted beneath the western margin of North America in Late Cretaceous and Paleogene time. Displacement parameters at a location for Baja California (Engebretson and others, 1985, Table 5) during this time interval for the Pacific plate, Farallon plate, and Kula plate when it might have been present, are listed in Table 3. Obliquity was calculated assuming an orientation of 345° for the North American margin. This azimuth is parallel to the present trend of the Cretaceous Sierra Nevada batholith, which is inferred, on the basis of modern examples, to have been approximately parallel to the ancient convergent plate margin (A. S. Jayko and M. C. Blake, Jr., unpub. data). Tangential displacements associated with Pacific plate motion after 28 Ma (Table 3) are estimated for the present trend ($\simeq 320°$) of the San Andreas fault (Engebretson and others, 1985). Jarrard (1986a) noted that there are active arc-parallel, strike-slip faults behind one-half of all modern subduction zones and that two-thirds of subduction zones with overriding continental crust have active arc-parallel, strike-slip faults. These faults tend to form 100 to 300 km inland from the trench (Fitch, 1972; Jarrard, 1986a), near or within the magmatic arc where the overriding plate is weakest (Beck, 1983).

The displacements of forearc slivers between the trench and strike-slip fault depend on the obliquity of plate convergence, the degree of interplate coupling, and the strength of the overriding plate (Beck, 1986; Jarrard, 1986a, 1986b). Jarrard (1986a, Table 5) calculated rates for the displacements of modern forearc slivers based on slip vector residuals, the differences between observed slip vector azimuths of shallow thrusting earthquakes and the expected azimuths from known plate motions. Using these data, A. S. Jayko and M. C. Blake, Jr. (unpub. data) estimated ancient strike-slip rates of allochthonous terranes within the Franciscan Complex. They assumed that the percentage of the total convergence rate for a given obliquity that is partitioned into arc-parallel strike-slip displacement at present-day subduction zones (Jarrard, 1986a) is equivalent to the percentage of reconstructed convergence rates (Engebretson and others, 1985) given to past strike-slip displacement. Adopting this approach, we have selected 15% partitioning of total convergence to arc-parallel strike-slip displacement for the Farallon plate, based on its average obliquity, and 50% partitioning for the Kula plate. These

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values for the Farallon and Kula plates are consistent with modern examples (Jarrard, 1986a), and are similar to values for the Ecuador and Colombia subduction zone and the Sumatra subduction zone, respectively. For the Pacific plate, the tangential displacement of the site (Table 3) is determined directly from the component of motion parallel to the plate margin.

The total tangential displacement due to strike-slip faulting since mid-Cretaceous time is 2,900 km for Pacific and Farallon plate motion, and 4,000 km for Pacific and Farallon + Kula plate motion (Table 3). These estimates are either approximately equivalent to, or greater than, the northward displacement inferred from the paleomagnetic data for the Cedros and San Benito chert during the same time interval. The exact timing of the northward translation of the Western Baja terrane within this time interval and whether or not the Kula plate was involved in the translation remain uncertain. This northward translation is also consistent with the suggestion of Karig (1980) that because the San Benito blueschists are too close to the trench and too old to have been related to the most recent subduction configuration, their location and exposure are most easily explained by arc-parallel, strike-slip faulting and related processes associated with earlier oblique subduction. A similar estimate of northward translation (≃2,500 km) has been made for the Salinian terrane of California (Champion and others, 1984) based on paleomagnetic results for the Upper Cretaceous Pigeon Point Formation and Paleocene strata at Point San Pedro. The results of our study also imply that other rocks of the Franciscan Complex could have been remagnetized prior to structural dismemberment (Tarduno and others, 1986, p. 1427) and that their remanent magnetizations could also reflect subsequent arc-parallel, northward displacement along the western margin of North America during Cenozoic time.

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